

From Hubble Action to Horizon Entropy: One Cosmological Hierarchy in Two Dimensional Bases

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Abstract. Adjoining the Hubble parameter to c , $\hbar$ and G admits exactly one dimensionless combination, $\Pi_0 = \hbar G H_0^2 / c^5 \approx 1.4 \times 10^{-122}$, and its reciprocal $N = \Pi_0^{-1} \approx 7.2 \times 10^{121}$ organises the large numbers of cosmological thermodynamics. The quantity $\mathcal{A}_H = M c^2 / H_0$, with M the mass of a critical-density Hubble sphere, carries the dimension of action, and counted in quanta of $\hbar$ gives $\mathcal{N}_H = \mathcal{A}_H / \hbar = \frac{1}{2} N \approx 3.6 \times 10^{121}$. Constructions that look independent then evaluate to fixed multiples of the same invariant: the Bekenstein bound for the critical sphere and the holographic area entropy both give $\pi N k_B \approx 2.3 \times 10^{122} k_B$, and the Gibbons–Hawking entropy gives $(\pi / \Omega_\Lambda) N k_B$; the spread between 10^{121} and 10^{122} is carried entirely by the prefactors. Such agreement is dimensional kinship, and what it identifies is a capacity scale. The thermodynamic inventory, computed with the statistical factors per particle stated in advance, lies far beneath it, at $7 \times 10^{89} k_B$ for the diffuse non-dark sector and near $10^{104} k_B$ once black holes are included. A Clifton–Ellis–Tavakol construction, in which gravitational entropy is sourced by the Weyl departure from homogeneity and vanishes in the FLRW limit, is bounded above by the same capacity, $S_{\text{CET}} \leq \pi N k_B$, the bound attained only if the full critical-sphere energy is assigned to the free field at the horizon temperature; the black-hole census suggests the integral sits near 10^{-18} of that limit today. The capacity of the evolving Hubble sphere grows at the fractional rate $2/t$, freezing as dark energy dominates. The 10^{-122} of the cosmological hierarchy and the 10^{122} of the horizon capacity are the same invariant read from opposite sides.

1 Buckingham Invariant

The constants c , $\hbar$ and G have dimension vectors spanning the mechanical space (M, L, T) ; their dimensional matrix has determinant -2 and full rank, which is the linear-algebraic licence for the natural-unit choice $c = \hbar = G = 1$ [1], [2]. Adjoining the Hubble parameter, of dimension T^{-1} , leaves the rank at three, so by Buckingham’s Π -theorem exactly one dimensionless combination exists,

$$\Pi_0 = \frac{\hbar G H_0^2}{c^5} = (H_0 t_P)^2 \approx 1.4 \times 10^{-122}, \quad (1)$$

with $t_P = \sqrt{\hbar G / c^5}$ the Planck time and $H_0 \approx 2.2 \times 10^{-18} \text{ s}^{-1}$. The hierarchy cannot be removed by a change of basis, only relocated: in Planck units it appears as $H_0^2 \sim 10^{-122}$, in a Hubble-adapted basis $c = \hbar = H_0 = 1$ as $G \sim 10^{-122}$, and through the de Sitter relation $H^2 = \Lambda c^2 / 3$ as $\Lambda \ell_P^2 \sim 10^{-122}$ [1]. Its reciprocal,

$$N \equiv \Pi_0^{-1} = \left(\frac{R_H}{\ell_P} \right)^2 \approx 7.2 \times 10^{121}, \quad (2)$$

is the Hubble-horizon area in Planck units, with $R_H = c / H_0 \approx 1.4 \times 10^{26} \text{ m}$ and $\ell_P = \sqrt{\hbar G / c^3}$.

This paper traces the recurring 10^{121} – 10^{122} of cosmological entropy bounds to N , computes the thermodynamic inventory that sits beneath those bounds in the same unit k_B , and bounds the Clifton–Ellis–Tavakol gravitational-entropy integral by the same capacity. Alongside R_H , the comoving radius of the observable universe, $R_{\text{obs}} \approx 4.4 \times 10^{26} \text{ m}$ with volume $V \approx 4 \times 10^{80} \text{ m}^3$, bounds the particle inventory; the two radii serve those two distinct purposes throughout.

2 Hubble action & quantum count

Dimensional analysis fixes the combination of H_0 and G carrying the dimension of density only up to a pure number, $\rho \propto H_0^2 / G$; the number is supplied by the Friedmann equation, whose flat solution sets $\rho_c = 3H_0^2 / 8\pi G$, the $3/8\pi$ owed to geometry rather than to dimensions. The mass this density assigns to the Hubble sphere,

$$M = \rho_c \cdot \frac{4}{3} \pi R_H^3 = \frac{c^3}{2GH_0} \approx 9.2 \times 10^{52} \text{ kg}, \quad (3)$$

with the $\frac{1}{2}$ likewise geometric, is the mass adopted throughout. From it, form

$$\mathcal{A}_H = \frac{Mc^2}{H_0} = McR_H \approx 3.8 \times 10^{87} \text{ J s}, \quad (4)$$

a quantity whose dimension, energy over inverse time, is that of action, and which is called here the *Hubble action*: the mass-energy of the critical sphere accumulated over one Hubble time. Since $\hbar$ is the quantum of action, the natural dimensionless measure of $\mathcal{A}_H$ is its quantum count,

$$\mathcal{N}_H = \frac{\mathcal{A}_H}{\hbar} = \frac{Mc^2}{\hbar H_0} = \frac{c^5}{2\hbar G H_0^2} = \frac{1}{2}N \approx 3.6 \times 10^{121}. \quad (5)$$

The count is a capacity-type quantity. A Shannon entropy would require an ensemble and a probability distribution over states, which Eq. (5) does not supply, so no conversion to bits is made here; particle-based information estimates [5, 6], which assign content to the particle inventory, concern a different and much smaller object.

The energy $\hbar H_0$ appearing in the count is a Hubble-scale quantum, to be distinguished from the Planck energy $E_P = \sqrt{\hbar c^5/G}$. The two names are kept to two duties: $\mathcal{A}_H$ is called an action because joule-seconds is the dimension it carries, while capacity describes the role its count and the bounds of Sec. 3 play.

Action and angular momentum share a dimension, and $\hbar$ is the quantum of both, so the count carries a second, mechanical reading worth a moment: $\mathcal{A}_H = McR_H$ is the angular momentum of the critical sphere's mass transported at the causal limit around the Hubble radius, $L_{\max} = \hbar \mathcal{N}_H \approx 3.6 \times 10^{121} \hbar$. A ladder of spins in the same unit sets its scale. A thousand children's tops at 10 rad s^{-1} carry $\sim 10^{34} \hbar$; the Solar System $3 \times 10^{77} \hbar$; the Milky Way $2 \times 10^{101} \hbar$; and a random assembly of the $N_g \sim 2 \times 10^{12}$ Milky-Way-like galaxies, with spin directions uncorrelated [?], carries the statistical residual $\sqrt{N_g} L_{\text{MW}} \approx 2.8 \times 10^{107} \hbar$, fourteen orders beneath the ceiling. The ladder is dimensional arithmetic and nothing besides: $\mathcal{A}_H$ is introduced here as an action scale, no claim of global cosmic rotation follows from it, and the role the count plays in what follows is that of a capacity, the scale against which the entropy bounds of the next section and the inventory beneath them are measured.

3 One invariant behind several horizon capacities

Several constructions of a cosmological entropy scale, arrived at by apparently different routes, evaluate to fixed multiples of the one invariant, and Table 1 collects them before any is derived, so that the spread of the numbers is visible at the outset: the count $\mathcal{N}_H$ sits at 10^{121} and the entropy bounds at 10^{122} , the whole difference carried by the prefactors $\frac{1}{2}$, π and π/Ω_Λ , the factor 2π between count and bound being the circumference of the quantum.

Setting a dimensionless count beside quantities carrying k_B asks for a word on units. Firstly, the basis $c = \hbar = G = 1$ absorbs the mechanical dimensions; temperature remains, Θ being a fourth base dimension whose lone carrier is k_B , exactly as the temperature row of a dimensional matrix pivots on k_B alone, so adjoining the row Θ and the column k_B preserves full rank and the completion $k_B = 1$ is licensed. Ontologically, k_B is a translation between units of measurement, kelvin into joule, of the same standing as the conversion of electron-volts into joules, a relic of temperature having received its own base unit before being recognised as a mean energy, so that setting $k_B = 1$ amounts to measuring temperature in energy.

Shannon entropy needs no such constant at all, being a pure logarithmic count over probabilities, dimensionless in nats, with division by $\ln 2$ giving bits as an exact change of logarithm base; k_B is required only for the return journey, from the count description back to Clausius entropy in J K^{-1} and temperature in kelvin. In the completed basis every entropy in the table is such a count. The physical assumption enters elsewhere: reading the count $\mathcal{N}_H$ as an entropy at all is an identification the dimensional analysis does not supply. Two cautions attend the choice. Natural-unit bases are conventions rather than unique, since several consistent completions exist and which constant carries the 10^{-122} depends on the basis chosen [1]; and the quantum itself may be taken as $\hbar$ or h , a convention of the same kind that relocates factors of 2π between count and prefactor, as the $(\hbar c)^3$ of Appendix A against the $\hbar$ of Eq. (5) illustrates. This paper counts in $\hbar$ and keeps k_B on display in the table as the dictionary between the thermodynamic description in J K^{-1} and the count description in nats.

Table 1: Routes to the capacity scale, as multiples of $N = \Pi_0^{-1} \approx 7.2 \times 10^{121}$. With $k_B = 1$ every entry is a dimensionless count in nats; k_B is displayed to keep the thermodynamic unit visible. The limit of the final row is derived in Sec. 6.

Route	Expression	$\times N$	Value
Hubble-action count	$Mc^2/\hbar H_0$	$\frac{1}{2}$	3.6×10^{121}
Bekenstein bound	$2\pi k_B ER/\hbar c$	π	$2.3 \times 10^{122} k_B$
Holographic area	$\pi k_B R_H^2/\ell_P^2$	π	$2.3 \times 10^{122} k_B$
Gibbons–Hawking	$3\pi k_B/\Lambda \ell_P^2$	π/Ω_Λ	$3.3 \times 10^{122} k_B$
CET upper limit	$Mc^2/k_B T_H$	π	$2.3 \times 10^{122} k_B$

The Bekenstein bound [7] for a system of energy E within radius R , $S \leq 2\pi k_B ER/\hbar c$, evaluated for the critical sphere $E = Mc^2$, $R = R_H$, gives

$$S_{\text{Bek}} = 2\pi \mathcal{N}_H k_B = \pi N k_B \approx 2.3 \times 10^{122} k_B, \quad (6)$$

and the holographic area entropy of the same sphere is $\pi k_B R_H^2/\ell_P^2 = \pi N k_B$ by Eq. (2); for the critical Hubble sphere the two coincide exactly, the critical density being precisely the density at which the energy bound and the area bound meet. The Gibbons–Hawking entropy, [3] of the de Sitter horizon, $3\pi k_B/(\Lambda \ell_P^2)$, joins them through $\Lambda \ell_P^2 = 3\Omega_\Lambda \Pi_0$. All the routes reduce algebraically to N , so their shared magnitude is dimensional kinship rather than the concordance of independent measurements; what the kinship identifies is a capacity scale, with no implication that the thermodynamic entropy of the universe saturates it.

In summary form, the 10^{-122} of the cosmological hierarchy and the 10^{122} of the horizon capacity are the same invariant read from opposite sides, $S \sim \pi k_B/\Pi_0$ against $\Lambda \ell_P^2 = 3\Omega_\Lambda \Pi_0$.

4 Capacity, and Thermodynamic inventory

Beneath the capacity sits the entropy the universe actually holds. The statistical mechanics enters through one number per species, the entropy carried per particle, stated first because everything downstream is a multiplication by it. For a boson gas in equilibrium the entropy per photon is a pure number,

$$s_\gamma = \frac{S_\gamma}{N_\gamma k_B} = \frac{2\pi^4}{45 \zeta(3)} \approx 3.6, \quad (7)$$

the ratio of the entropy density $\frac{4}{3}aT^3$ to the number density bT^3 , with $\zeta(3)$ arising from the Bose–Einstein integral (Appendix A).

For the relativistic Fermi–Dirac neutrino background the corresponding factor is $s_\nu \approx 3$ [8, 9], reduced from the photon value by the 7/8 family of fermionic integrals.

For the non-relativistic, distinguishable baryons the classical count gives $s_b \approx \ln N_b \approx 184$ per particle. Number densities are fixed by the CMB temperature $T_\gamma = 2.725$ K, the decoupling-cooled neutrino temperature $T_\nu = (4/11)^{1/3} T_\gamma \approx 1.95$ K [10, 11], and the baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$.

Table 2 overleaf is then the whole of the diffuse non-dark accounting.

Two remarks tie the table to familiar quantities. The specific entropy, the photon count per baryon, is

$$S_{\gamma b} = \frac{n_\gamma}{n_b} = \frac{1}{\eta} \approx 1.7 \times 10^9, \quad (8)$$

in the sense of Prigogine [12, 13], and the entropy per baryon that follows from the table, $S_{\text{bary}}/N_b \approx 7 \times 10^9 k_B$, is $s_\gamma S_{\gamma b}$ to within the neutrino contribution, as it must be. Second, the photon internal energy obeys the black-body identity $T_\gamma S_\gamma = \frac{4}{3}U_\gamma$, with $U_\gamma = aT_\gamma^4 V = 1.7 \times 10^{67}$ J and $T_\gamma S_\gamma k_B = 2.2 \times 10^{67}$ J. The table concerns the diffuse sector: admitting stellar and supermassive black holes lifts the inventory to the census value of roughly $3 \times 10^{104} k_B$, dominated by supermassive black holes [14]. Set against the capacity $\pi N k_B \approx 2.3 \times 10^{122} k_B$, the diffuse sector lies thirty-two orders beneath, the census eighteen; the difference between an entropy and a bound is capacity unused, as an empty disk conceals no hidden files, so the separation calls for no missing component. Sec. 6 places a limit on the gravitational-entropy integral that sits between content and capacity.

5 Chemical potentials, and particle bookkeeping

The accounting rests on the second law in the form

$$k_B T dS = dU + p dV + \mu dN, \quad (9)$$

with S in units of k_B (nats), and with the radiation equation of state $p = \rho/3$ following from the isotropy of the energy-momentum tensor of a perfect photon fluid. The treatment sets $\mu = 0$ for photons and yet computes particle-number changes dN , and the licence for both at once is exact. Expressing $U = aT^4 V$ through $N = bT^3 V$

Table 2: The diffuse non-dark inventory of the observable universe ($V = 4 \times 10^{80} \text{ m}^3$). Internal energies are thermal and enter the chemical-potential accounting of Sec. 5; the baryon rest-mass energy $M_b c^2 \approx 1.5 \times 10^{70} \text{ J}$ is excluded from the thermodynamic accounting by the baryon chemical potential.

Species	Statistics	$n [\text{m}^{-3}]$	N	s/k_B per particle	$S [k_B]$	$U_{\text{th}} [\text{J}]$
photons	Bose–Einstein	4.1×10^8	1.6×10^{89}	3.60	5.9×10^{89}	1.7×10^{67}
neutrinos	Fermi–Dirac	1.1×10^8	4.5×10^{88}	≈ 3	1.3×10^{89}	3×10^{66}
baryons	classical	2.5×10^{-1}	9.9×10^{79}	$\ln N_b \approx 184$	1.8×10^{82}	5.5×10^{57}
total					$S_{\text{bary}} \approx 7.3 \times 10^{89}$	$U \approx 2.0 \times 10^{67}$

gives $U = a b^{-4/3} N^{4/3} V^{-1/3}$, whence the energy cost of one more photon at fixed volume is

$$\begin{aligned} \left(\frac{\partial U}{\partial N} \right)_V &= \frac{4}{3} a b^{-4/3} N^{1/3} V^{-1/3} \\ &= 1.36 \times 10^{-22} \text{ J} = 3.6 k_B T_\gamma, \end{aligned} \quad (10)$$

which equals T_γ times the entropy per photon of Eq. (7). A photon added at fixed volume brings energy $3.6 k_B T$ and entropy $3.6 k_B$, so the full chemical potential vanishes identically,

$$\mu_\gamma = \left(\frac{\partial U}{\partial N} \right)_V - T \left(\frac{\partial S}{\partial N} \right)_V = 3.6 k_B T - 3.6 k_B T = 0. \quad (11)$$

Photon number is a dependent variable, free to change at zero thermodynamic cost, and Eq. (9) with $\mu_\gamma = 0$ remains exact while dN_γ is tracked as bookkeeping. The same holds for the neutrino background after decoupling. For baryons the opposite limit applies: baryon number is conserved and the chemical potential is dominated by the rest energy, $\mu_b \approx m_b c^2$, which is why the rest-mass energy $M_b c^2 \approx 1.5 \times 10^{70} \text{ J}$, though eight hundred times the photon energy, does not enter the entropy budget: the $\mu_b dN_b$ term cancels it from $k_B T dS$, leaving only the thermal $U_b = \frac{3}{2} N_b k_B T_b = 5.5 \times 10^{57} \text{ J}$ to participate.

The CMB anisotropy sets the scale of thermodynamic response. With $dT/T \approx 3.7 \times 10^{-6}$ [15, 16] and $S \propto T^3$ for radiation, $dS/S = 3 dT/T \approx 10^{-5}$, and the photon-number change $dN_\gamma/N_\gamma = dS/S$, licensed by $\mu_\gamma = 0$, exceeds the Poisson expectation $N_\gamma^{-1/2} \approx 2.5 \times 10^{-45}$ by forty orders of magnitude, the sense in which the fluctuations are dynamical rather than statistical; structure acts as a photon multiplier, re-radiating absorbed starlight as many infrared quanta at zero chemical cost.

6 Bounding gravitational-entropy integral

Clifton, Ellis and Tavakol [4] assign the free gravitational field an effective thermodynamics,

$$T_g dS_{\text{grav}} = dU_{\text{grav}} + p_g dV, \quad (12)$$

sourced by the Bel–Robinson super-energy in place of the ordinary energy-momentum tensor, with the equation of state fixed by the Petrov class of the spacetime: radiation-like, $p_g = \rho_g/3$, for the type N fields of gravitational radiation, and pressureless, $p_g = 0$, for the type D Coulomb-like fields, the class of the Lemaître–Tolman–Bondi geometries. In this construction the gravitational energy density measures the departure from homogeneity,

$$8\pi\rho_g \propto \frac{|m - \frac{4\pi}{3} R^3 \rho|}{R^3}, \quad (13)$$

the difference between the enclosed mass and the mass a homogeneous distribution would place in the same areal volume; it vanishes in the FLRW limit and grows as structure develops. The gravitational entropy of the Hubble domain is accordingly the spatially resolved integral, schematically

$$S_{\text{CET}} \propto \int_{\nu_H} \frac{\rho_g(r, t)}{T_g(r, t)} dV, \quad (14)$$

its value dependent on the chosen density profile, bang-time function, radial domain and CET normalisation, none of which are fixed here.

What can be fixed is the ceiling of Eq. (14). If the full critical-sphere energy were assigned to the free field, $\rho_g c^2 V \rightarrow M c^2$, at the horizon temperature

$$T_H = \frac{\hbar H_0}{2\pi k_B} \approx 2.7 \times 10^{-30} \text{ K}, \quad (15)$$

then with $p_g = 0$,

$$\begin{aligned} S_{\text{CET}}^{\text{max}} &= \frac{M c^2}{k_B T_H} = 2\pi \mathcal{N}_H k_B \\ &= \pi N k_B \approx 2.3 \times 10^{122} k_B, \end{aligned} \quad (16)$$

the Bekenstein and holographic value of Table 1. Writing

$$S_{\text{CET}} = f_{\text{CET}} \pi N k_B, \quad 0 \leq f_{\text{CET}} \leq 1, \quad (17)$$

records how far the integral lies beneath its limit: the FLRW limit is $f_{\text{CET}} = 0$, a maximally condensed configuration $f_{\text{CET}} = 1$, the partially structured universe between. The black-hole census, $\sim 3 \times 10^{104} k_B$ [14], supplies an independently estimated scale for realised gravitational entropy, and read against the limit it suggests $f_{\text{CET}} \sim 10^{-18}$ today. Nothing more is claimed here than the limit itself: Eq. (16) bounds the integral of Eq. (14) from above, evaluating that integral would require a specified density profile, bang-time function, radial domain and normalisation, and no such evaluation is attempted.

7 Evolution, and conclusion

In this section H denotes the time-dependent Hubble parameter, $H = H(t) = \dot{a}/a$, and $H_0 = H(t_0)$ its present value. Through the radiation and matter eras $H(t) = \alpha/t$ with $\alpha = \frac{1}{2}$ and $\frac{2}{3}$ respectively, so $-\dot{H}/H = 1/t$; in the dark-energy era H stabilises at $H_\infty = H_0 \sqrt{\Omega_\Lambda}$ and $\dot{H} \rightarrow 0$ [17]. Two readings of M give two laws for the count: a fixed material parcel of constant mass-energy has $\mathcal{N}_H = Mc^2/\hbar H \propto H^{-1}$, growing at $\dot{\mathcal{N}}_H/\mathcal{N}_H = 1/t$, while the evolving critical Hubble sphere, whose mass $M(t) = c^3/2GH(t)$ itself scales, has

$$\mathcal{N}_H(t) = \frac{c^5}{2\hbar GH(t)^2} \propto \frac{1}{H^2}, \quad \frac{\dot{\mathcal{N}}_H}{\mathcal{N}_H} = \frac{2}{t}, \quad (18)$$

the law of the horizon area itself, so the evolving sphere is the reading that tracks horizon entropy; with the concordance age satisfying $t_0 H_0 \approx 1$, its rate today is $\approx 2H_0 \approx 4.4 \times 10^{-18} \text{ s}^{-1}$, freezing at $Mc^2/(\hbar H_\infty)$ once dark energy dominates.

The hierarchy then stands complete. The diffuse inventory holds $7 \times 10^{89} k_B$, the census with black holes $3 \times 10^{104} k_B$, the horizon capacity $\pi N k_B \approx 2.3 \times 10^{122} k_B$, eighteen orders above the census and growing at $2/t$; the census sits near 10^{-18} of the bound on the gravitational-entropy integral; and the size of the capacity is the reciprocal of the smallest number in physics, the 10^{-122} of the cosmological hierarchy and the 10^{122} of the horizon capacity being the same invariant read from opposite sides.

A Bose–Einstein integral & $\zeta(3)$

The number density of a photon gas integrates the Bose–Einstein distribution over states, $n_\gamma = \int_0^\infty g(\epsilon) (e^{\epsilon/k_B T} - 1)^{-1} d\epsilon$ with $g(\epsilon) = 8\pi\epsilon^2/(hc)^3$; substituting $x = \epsilon/k_B T$,

$$n_\gamma = \frac{8\pi(k_B T)^3}{(hc)^3} \int_0^\infty \frac{x^2}{e^x - 1} dx = \frac{16\pi\zeta(3)(k_B T)^3}{(hc)^3}. \quad (19)$$

The $\zeta(3)$ falls out as a convergent cumulative sum over $1/n^3$, and it is worth watching it do so. Expanding the Bose factor geometrically, $(e^x - 1)^{-1} = \sum_{n \geq 1} e^{-nx}$, each term integrates through the gamma function: with $u = nx$,

$$\int_0^\infty x^2 e^{-nx} dx = \frac{1}{n^3} \int_0^\infty u^2 e^{-u} du = \frac{\Gamma(3)}{n^3} = \frac{2}{n^3}, \quad (20)$$

so the integral is twice the cumulative sum $\sum_{n \geq 1} n^{-3}$, whose partial sums climb monotonically to $\zeta(3) = 1.2020569 \dots$. The Fermi–Dirac integral gives $\frac{3}{2}\zeta(3)$ *mutatis mutandis*, whence $n_\nu = \frac{3}{11} n_\gamma$ per the decoupling temperature ratio, and the entropy-per-particle factors of Sec. 4 follow as ratios of the corresponding $\zeta(4)$ -type energy integrals to these number integrals.

As narrative only: this same $\zeta(3)$ is the factor through which the numerical curiosity of [1] brings the Ramanujan constant into registration with G as computed through H_0 , entering there exactly as here, by the cumulative $1/n^3$ sum; no mechanism being implied.

References

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